

AIM Technical Questions

Multi-task molecular property prediction on QM9 — pretrained Uni-Mol vs a from-scratch MPNN, with AIM, PCGrad and linear scalarisation

Fifty technical, mathematical and theoretical questions on this project, written at the level of questions a professor is likely to ask after the talk. Each question is followed by the key points a strong answer covers. Questions marked in colour concern issues found in the current pipeline: expect them, and answer them before they are asked.

Sections: A problem setting and data · B Uni-Mol representation and architecture · C pretraining and transfer · D the MPNN reference backbone · E multi-task optimisation theory · F AIM in detail · G evaluation, statistics and validity.

A. Problem setting, data and targets

Whether the candidate understands what is being predicted, and why these three targets were chosen.

Q1. μ is intensive; U_0 and U are extensive. State extensivity formally: for a system split into non-interacting fragments A and B, $E(A \cup B) = E(A) + E(B)$. Which of your two readouts — Uni-Mol's [CLS] attention pooling and the GNN's Set2Set — can represent an extensive quantity exactly, and what does the answer imply about the hypothesis class you chose for the energy tasks?

Key points: Attention/Set2Set produce convex combinations of atom states, so outputs stay bounded by the per-atom range and cannot grow linearly with N ; sum pooling can. This is a representational limitation, not an optimisation failure.

Q2. [PIPELINE AUDIT] The standard deviation of U_0 on QM9 is ≈ 1081 eV and you train L1 on z-scored targets. Convert a 40 eV MAE into normalised units and explain why the training loss can look converged while the model is chemically useless.

Key points: $40/1081 \approx 0.037$ in normalised units. The loss is dominated by the huge scale, so the optimiser has no pressure left in the region that matters chemically (~ 0.04 eV).

Q3. [PIPELINE AUDIT] A least-squares fit of per-element reference energies (6 parameters) reaches 0.87 eV on your test split, while your networks reach 37–55 eV. What does this tell you about what the networks have learned, and what is the correct reformulation of the target?

Key points: They have not even learned composition. Predict the residual $U_0 - \sum_Z n_Z \epsilon_Z$ (atomisation-like) and/or use a size-extensive readout; then the network models bonding rather than atom counting.

Q4. $U = U_0 + a$ thermal correction with mean 0.23 eV, so $\cos(g_{U_0}, g_U) \rightarrow 1$. If $y_2 = y_1 + \epsilon$ with $\|\epsilon\|$ small, what is the effective rank of the task-loss Jacobian, how many tasks do you really have, and what is the dimension of the Pareto front?

Key points: Effectively two tasks; the energy direction is duplicated and therefore double-weighted under LS. The Pareto front collapses along the U_0 - U direction.

Q5. You use a random split, not a scaffold split. Which quantity is your test MAE an unbiased estimate of, where does the i.i.d. assumption break for GDB-derived molecules, and in which direction is the bias?

Key points: It estimates in-distribution error on GDB-9 enumerated molecules; near-duplicate scaffolds appear on both sides, so it is optimistic relative to any prospective (new-scaffold) use.

Q6. Normalisation statistics come from the primary split only. Derive how a mis-scaled target changes the relative magnitudes of task gradients, and hence the cosine-based intervention itself.

Key points: Scaling y_t by c scales L_t and g_t by c ; cosines are invariant but $\|g_t\|$ and thus the LS sum, L_{mag} and L_{prog} are not. Task weighting is implicit in the normalisation choice.

Q7. Why L1 rather than MSE for QM9 regression, and how does the L1 subgradient behave as errors approach zero? What does that imply for gradient norms and for cosine estimates late in training?

Key points: L1 gives a constant-magnitude subgradient, robust to outliers but noisy near zero; gradient norms do not decay as the fit improves, so conflict statistics stay noisy at convergence.

B. Uni-Mol: representation and architecture

The core of the project: what the pretrained encoder actually computes.

Q8. Prove that the encoder is invariant to rigid motions, given that coordinates enter only through D . Then explain why it is also invariant to reflection, and name a property class that this makes unlearnable.

Key points: $D(RQ + b) = D(R)$ for $Q \in O(3)$, which includes $\det = -1$. Enantiomers share a distance matrix, so any chirality-dependent property (optical rotation, differential binding) is out of reach.

Q9. The pair channel applies a per-pair-type affine map $\tilde{d} = a_{\tau} d + b_{\tau}$ before 128 Gaussians $p_k = N(\tilde{d}; \mu_k, \sigma_k)$. Why make a, b depend on the atom-pair type, and what function class does the Gaussian basis span?

Key points: It lets the network place the basis where each element pair's chemistry lives (C-H ≈ 1.1 Å vs C-C ≈ 1.5 Å). The basis spans a smooth RBF space; any continuous radial function is approximated, with smoothness set by σ .

Q10. Attention is $\text{softmax}(QK^T/\sqrt{d} + q)$. Show that adding the pair bias is equivalent to imposing a prior on attention logits, and explain how distance-decaying attention arises.

Key points: The bias shifts logits multiplicatively in probability space (softmax of a sum = product of exponentials), i.e. a learned geometric prior over which atoms may attend to which.

Q11. The pair representation is updated by $q_{ij} \leftarrow q_{ij} + Q_{iK_j}^T/\sqrt{d}$. Give the compute and memory scaling in the number of atoms and explain the consequence for batching padded molecules.

Key points: $O(N^2)$ per layer for pairs ($O(N^2 \cdot H)$ memory), so cost is set by the largest molecule in the batch; padding to T wastes compute quadratically.

Q12. Compare [CLS] pooling with sum pooling. Prove sum pooling can express an extensive target exactly, and state what attention pooling can guarantee instead.

Key points: $\sum_i f(h_i)$ grows linearly with the number of atoms; a convex combination $\sum a_i h_i$ with $\sum a_i = 1$ stays in the convex hull, so extensivity must be recovered indirectly through molecule-size features.

Q13. [PIPELINE AUDIT] Your collator marks position 0 with the id Uni-Mol reserves for [PAD], while padded positions carry the id for [CLS]. Derive the two consequences for the attention computation, and design a test that detects it from model outputs alone.

Key points: The pooled position is excluded from keys; padded slots are treated as real tokens at distance 0 from everything. Test: run one molecule alone and in a batch with a large molecule — predictions should be identical, and are not.

Q14. [PIPELINE AUDIT] Pair types are encoded $e_{ij} = t_i \cdot |V| + t_j$. Show this is injective only if $|V|$ is at least the vocabulary size, state why the embedding table has 961 rows, and explain what happens with $|V| = 58$ instead of 31.

Key points: $961 = 31^2$. With a larger multiplier, ids collide with, or index into, rows belonging to different element pairs, so the learned per-pair affine map is silently mis-selected.

Q15. [PIPELINE AUDIT] Suppose token ids do not match those used during pretraining. Explain why fine-tuning only the last two of fifteen blocks cannot repair this, using an argument about where the mismatch enters the computation.

Key points: The mismatch is at the embedding and pair-kernel level, which is frozen; the top blocks receive systematically wrong features and can only apply a limited correction — comparable to linear probing on corrupted inputs.

Q16. The encoder has 512 hidden units across 64 heads, so head dimension 8. Discuss the trade-off: what does the rank of each head's value projection bound, and why might many small heads be preferable here?

Key points: Each head can move at most an 8-dimensional subspace; many heads give many specialised, low-rank geometric relations — well matched to a pair-bias attention where each head can encode a distance regime.

Q17. Uni-Mol is invariant, not equivariant. Which quantities need equivariance, and what concretely would you change to predict the dipole vector rather than its norm?

Key points: Vector/tensor targets (dipole vector, forces, polarisability). You need vector features that rotate with the input — e.g. predict per-atom charges and form $\sum q_i \mathbf{r}_i$, or move to an equivariant architecture.

C. Pretraining and transfer

Why a pretrained encoder should help, and under which assumptions.

Q18. Uni-Mol's pretraining includes coordinate denoising. Relate this objective to score matching, and argue why learning $\nabla \log p(\mathbf{x})$ over conformations produces features relevant to energies.

Key points: Denoising estimates the score of the conformer distribution; since $p(\mathbf{x}) \propto \exp(-E(\mathbf{x})/kT)$ at equilibrium, the score is proportional to a force field, so denoising implicitly learns the local shape of the energy surface.

Q19. You fine-tune 6.3 M parameters on 4000 molecules. State the naive capacity concern, then give the argument for why this does not necessarily overfit catastrophically.

Key points: Parameter counting is the wrong measure; what matters is the distance travelled from the pretrained initialisation in an implicitly regularised, low-effective-dimension subspace under a small learning rate.

Q20. Define the hypothesis classes for linear probing, partial fine-tuning and full fine-tuning, and show they are nested. Where does trainable_layers = 2 sit and what does the nesting say about achievable training error?

Key points: $H_{\text{probe}} \subseteq H_{\text{partial}} \subseteq H_{\text{full}}$: training error is monotone non-increasing along the chain, while variance and feature distortion increase — the classic bias/variance ladder.

Q21. Explain feature distortion during fine-tuning (Kumar et al., 2022): why can fine-tuning underperform linear probing under distribution shift, and how does freezing lower layers mitigate it?

Key points: Early large gradients from a randomly initialised head distort good pretrained features before the head is sensible; freezing (or head warm-up with a small lr) protects them.

Q22. Uni-Mol was pretrained largely on force-field-generated conformers, while QM9 provides DFT-optimised geometries. Characterise this shift and its likely effect on transfer.

Key points: A systematic shift in bond lengths and angles of order hundredths of an Å, entering directly through the Gaussian distance basis; the effect is a covariate shift in the pair channel rather than in the atom tokens.

D. The GNN / MPNN reference backbone

The from-scratch baseline and what it isolates. Keep this brief in the talk, but be ready to defend it.

Q23. Your MPNN connects all atom pairs within 10 Å, which for QM9 is essentially a complete graph. What then remains as the real difference between your MPNN and a Transformer with a distance bias?

Key points: The message function: a distance-conditioned linear map $A(\mathbf{e}_{vw})\mathbf{h}_w$ with mean aggregation, versus content-based attention weights. The MPNN's mixing weights do not depend on the source atom's features.

Q24. The edge network maps 16 RBF features to a 64×64 matrix. Give the parameter count, and explain why a full matrix is more expressive than the scalar continuous filter used in SchNet.

Key points: $128 \times 4096 \approx 524 \text{ k}$ per block. A matrix can rotate and mix feature channels per pair; a scalar filter can only rescale them elementwise.

Q25. Message-passing networks are bounded by 1-WL on discrete graphs. What does that bound mean when edges carry continuous distances, and can your MPNN distinguish geometric graphs that 1-WL cannot?

Key points: With injective continuous edge attributes the colour refinement operates on a richer alphabet, so the discrete 1-WL bound is not the binding constraint; geometry, not topology, is the discriminating signal.

Q26. Why update node states with a GRU cell rather than $h \leftarrow h + m$? Answer in terms of gradient flow over T steps and of controlling how much neighbourhood information is admitted.

Key points: Gating gives a learned, bounded interpolation and mitigates vanishing/exploding gradients through the recurrence; it also lets an atom ignore uninformative messages.

Q27. Prove Set2Set is permutation invariant, and state whether it is extensive. How does it compare with the DeepSets universality result?

Key points: Attention weights are computed per element and summed, so any permutation gives the same result. It is a weighted average — not extensive. DeepSets shows sum-decomposition is universal for set functions.

Q28. [PIPELINE AUDIT] In your runs the node projection and the first message-passing block stayed at their random initialisation. What does a random linear projection preserve, and is this fatal for the baseline?

Key points: By Johnson-Lindenstrauss, random projections approximately preserve distances/inner products, so it acts as a fixed random feature map — a handicap and a confound, but not a catastrophe.

Q29. The MPNN trains 1.16 M parameters and beats the 6.3 M-trainable Uni-Mol on U_0 and U . Give at least two competing explanations and an experiment that separates them.

Key points: (i) Corrupted Uni-Mol inputs; (ii) explicit composition features (element one-hot, H counts) help an extensive target; (iii) more of the network is adapted. Fix the tokenisation and re-run, and ablate atom features.

E. Multi-task optimisation: theory

The mathematics the method rests on. Expect the sharpest questions here.

Q30. Define Pareto stationarity for the vector of task losses. Prove that, for convex losses, a minimiser of $\sum w_t L_t$ with all $w_t > 0$ is Pareto optimal, and explain what survives in the non-convex case.

Key points: If another point improved one task without harming others it would lower the weighted sum — a contradiction. Non-convex: only local/stationarity statements remain, and different w trace different local solutions.

Q31. State the MGDA problem: the minimum-norm element of the convex hull of task gradients. Write its KKT conditions and explain the relationship to PCGrad.

Key points: $\min \|\sum \alpha_t g_t\|^2$ s.t. $\alpha \geq 0, \sum \alpha = 1$. The solution is a common descent direction or a certificate of Pareto stationarity; PCGrad is a cheap heuristic aimed at the same conflict, without the optimality guarantee.

Q32. Derive the exact first-order condition under which one LS step increases task i 's loss, and interpret each term.

Key points: $\Delta L_i \approx -\eta(\|g_i\|^2 + \sum_{t \neq i} \langle g_i, g_t \rangle) > 0 \iff \sum_{t \neq i} \|g_t\| \cos \phi_{it} < -\|g_i\|$. Conflict alone is insufficient: a conflicting gradient must also be large enough.

Q33. PCGrad projects sequentially in a random task order. Construct a three-task example where two orders give different updates, and state what property is thereby lost.

Key points: With mutually conflicting gradients, projecting i onto j then k differs from k then j ; the map is order-dependent, so the update is not a well-defined function of $\{g_i\}$ — only its expectation over orders is.

Q34. For $T = 2$ with $\cos \phi < 0$, compute the PCGrad-corrected sum explicitly and show its inner product with each g_i is non-negative.

Key points: $g_1' = g_1 - (\langle g_1, g_2 \rangle / \|g_2\|^2) g_2$ is orthogonal to g_2 , and likewise for g_2' . Their sum has $\langle \cdot, g_i \rangle = \|g_i'\|^2 \geq 0$, so neither task is harmed to first order.

Q35. LS averages the gradients while PCGrad and AIM sum them — a factor T . Explain why Adam makes this almost irrelevant, and state precisely the invariance Adam does and does not have.

Key points: Adam divides by a running RMS, so it is invariant to a global constant rescaling of the gradient (up to ϵ and warm-up), but not to per-coordinate rescaling or to changes in gradient direction.

Q36. Why is the cosine an incomplete description of conflict? Define the magnitude-imbalance ratio $\kappa = \max\|g_t\|/\min\|g_t\|$ and give a case with harmless negative cosine and one with harmful positive cosine.

Key points: A tiny conflicting gradient is harmless; a large aligned-but-dominant gradient can still starve a task's progress in the directions it needs. Conflict is about the pair (angle, magnitude).

Q37. In $d \approx 6.3 \cdot 10^6$ dimensions, two independent random vectors have $\cos \approx 0$ with standard deviation $1/\sqrt{d} \approx 4 \cdot 10^{-4}$. Given that, what is the statistical meaning of a learned threshold at $\tau = -0.05$, and what would you need to measure to interpret it?

Key points: It is enormous relative to chance, so measured cosines are far from random — but τ is a threshold on cosines that were never logged. Without the empirical distribution of $\cos \phi_{ij}$, τ cannot be interpreted.

Q38. Gradients are estimated from mini-batches of 32. Is the expected mini-batch cosine equal to the full-batch cosine? Justify, and describe how batch size biases any conflict statistic.

Key points: No — cosine is a nonlinear function of the gradients, so by Jensen's inequality the estimate is biased; noise pushes cosines toward zero, and the bias shrinks as batch size grows.

Q39. Kurin et al. and Xin et al. report that tuned scalarisation matches specialised MTO methods. Design the experiment that would convincingly establish 'AIM > LS' on your setup: tuning budget, seeds, and statistical test.

Key points: Equal hyperparameter budget for LS (including task weights), ≥ 5 seeds, a pre-registered metric, paired tests across seeds plus bootstrap over test molecules, and reporting of effect size, not just a win.

F. AIM: the method in detail

Equations 1–6 and what your implementation actually does.

Q40. Analyse $w_{ij} = \sigma(k(\tau_{ij} - \cos \phi_{ij}))$ in the limits $k \rightarrow \infty$ and $\tau \rightarrow \pm 1$. Show PCGrad is a special case and state the practical benefit of a smooth gate.

Key points: $k \rightarrow \infty$ with $\tau = 0$ gives the indicator $1[\cos < 0]$, i.e. PCGrad; $\tau \rightarrow 1$ always projects, $\tau \rightarrow -1$ never does. Smoothness makes the update differentiable in τ , which is what allows the policy to be learned.

Q41. In Equation 2 the projection is applied with weight w_{ij} even when $\cos \phi_{ij} > 0$. Derive a condition on w under which $\langle g'_i, g_i \rangle < 0$ — that is, over-projection reverses task i 's own descent direction.

Key points: For two tasks, $\langle g'_i, g_i \rangle = \|g_i\|^2(1 - w \cos^2 \phi)$. It becomes negative once $w > 1/\cos^2 \phi$, which the sigmoid cannot reach for a single pair, but summing over j can.

Q42. Equation 5 sets $L_{\text{mag}} = (\|g_{\text{AIM}}\| - \sum_i \|g_i\|)^2$. Using the triangle inequality, show the target is attainable only in a degenerate case, and describe the pressure this puts on τ .

Key points: $\sum \|g_i\| \geq \|\sum g_i\|$ with equality iff all gradients are parallel and equally oriented. The term therefore pushes τ to project as little as possible — a bias toward inaction, not a norm-preserving constraint.

Q43. Equation 6 uses $L_{\text{prog}} = -\sum_i \alpha_i \langle g_{\text{AIM}}, g_i \rangle$ with $\alpha_i = L_i / \sum_j L_j$. Show it scales with $\|g\|^2$, explain why the same λ_p means different things on your two backbones, and propose a scale-free version.

Key points: Measured $L_{\text{prog}} \approx -2.8$ (Uni-Mol) vs -240 (MPNN): identical λ , very different regularisation. Normalise by $\sum \|g_i\|^2$ or use cosines instead of inner products.

Q44. [PIPELINE AUDIT] In your implementation the guidance losses are detached constants, so L_{guide} contributes no gradient to τ . Derive $\partial L_{\text{policy}} / \partial \tau_{ij}$, interpret its sign, and predict the fixed point.

Key points: Only L_{mag} and L_{prog} reach τ , both through w_{ij} ; since both reward smaller projections, τ drifts negative until the gate is nearly closed. That matches the observed $\tau \in [-0.25, -0.02]$ — and is a property of the regularisers, not evidence about the encoder.

G. Evaluation, statistics and validity

Where an ETH committee will press hardest: what your numbers can actually support.

Q45. You select the checkpoint by the mean validation MAE across three tasks measured in different units (D and eV). Show that this implicitly weights the tasks, and propose a scale-free selection criterion.

Key points: The eV-scale tasks dominate the mean by three orders of magnitude, so selection is effectively on energies alone. Use the mean of per-task normalised errors or of $\Delta\%$ against a fixed reference.

Q46. [PIPELINE AUDIT] For single-task runs, the same rule averages over two heads that were never trained. Explain the failure mode and how you corrected it.

Key points: The untrained heads' errors dwarf the trained one, so the argmin lands at an arbitrary early epoch. Correct rule: select on the trained task's own validation metric.

Q47. $\Delta m\% = (\text{MAE_STL} - \text{MAE})/\text{MAE_STL}$ is asymmetric under exchanging the two models. Demonstrate the asymmetry and propose a symmetric alternative.

Key points: Halving gives +50 %, doubling gives -100 %. A log-ratio $\log(\text{MAE}/\text{MAE_STL})$ is symmetric and additive, and averages sensibly across tasks.

Q48. You have one seed and observe differences of a few percent between methods. What can you claim, and how many seeds would you need to detect a 5 % effect at 80 % power?

Key points: With $n = 1$, nothing inferential — only description. The seed count follows from the seed-to-seed standard deviation, which is itself unmeasured; the honest first step is to estimate it with 3-5 seeds.

Q49. [PIPELINE AUDIT] Set out the baseline hierarchy an evaluation of extensive targets must report, and state what the 0.87 eV composition baseline implies about the ceiling of your current setup.

Key points: Mean predictor $\rightarrow$ composition (dressed-atom) baseline $\rightarrow$ STL $\rightarrow$ MTL. Any model above the composition baseline has not yet learned chemistry, so method comparisons on those targets are not yet meaningful.

Q50. Design the definitive experiment for 'does pretraining reduce the need for gradient intervention?'. Specify the measurements, the controls, and — in advance — which outcome would support and which would refute the hypothesis.

Key points: $\log \cos \phi_{ij}$ and $\|g_t\|$ per step; ablate trainable depth (2/6/15 blocks); full method $\times$ seed grid on corrected targets. Support: conflict rate and AIM's margin over LS both fall with pretraining and rise with depth. Refutation: conflict statistics comparable across backbones, with AIM's margin within seed noise.